

Worksheet: Propositional Logic & First-Order Logic

Foundations of Artificial Intelligence – SS 2026 *University of Bremen,
Institute for Artificial Intelligence*

Instructions: Written tasks () should be answered directly in this document or in a Markdown cell in the notebook. Programming tasks () belong in `logic_exercises.ipynb`. Estimated time: ~60 minutes.

Part 1 – Propositional Logic: Entailment & Satisfiability (~10 min)

1.1 Is the following entailment true or false? Justify your answer.

$$A \models A \vee B$$

1.2 Is the formula $(A \rightarrow B) \vee (B \rightarrow A)$ a tautology? Prove your answer using a truth table.

1.3 For each formula below, state whether it is **satisfiable**, **unsatisfiable**, or a **tautology**:

#	Formula	Satisfiable?	Unsatisfiable?	Tautology?
i	$A \wedge \neg A$			
ii	$A \vee \neg A$			
iii	$(A \rightarrow B) \wedge (B \rightarrow A)$			
iv	$\neg(A \vee B) \wedge A$			

1.4 Does $A \leftrightarrow B$ entail $A \rightarrow B$? Explain why.

1.5 Which of the following can be entailed from $A \wedge B$?

- (a) $A \vee B \vee C$
 - (b) $A \leftrightarrow B$
 - (c) None of the above
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Part 2 – Counting Models (~8 min)

Consider a vocabulary with four propositional symbols: A, B, C, D .
There are $2^4 = 16$ possible truth assignments in total.

For each sentence, count how many of the 16 models satisfy it:

(a) $(A \wedge B) \vee (B \wedge C)$

(b) $A \vee B$

(c) $(A \leftrightarrow B) \leftrightarrow D$

Part 3 – Validity and Unsatisfiability (~10 min)

For each sentence, decide whether it is **valid** (tautology), **unsatisfiable**, or **neither**.

Briefly justify each answer — a truth table or a short argument is fine.

	Formula	Valid / Unsat / Neither
a	$\text{Smoke} \rightarrow \text{Smoke}$	
b	$(\text{Smoke} \rightarrow \text{Fire}) \rightarrow$ $(\neg \text{Smoke} \rightarrow$ $\neg \text{Fire})$	
c	$\text{Smoke} \vee \text{Fire} \vee$ $\neg \text{Fire}$	
d	$((\text{Smoke} \wedge \text{Heat}) \rightarrow$ $\text{Fire}) \leftrightarrow$ $((\text{Smoke} \rightarrow \text{Fire}) \vee$ $(\text{Heat} \rightarrow \text{Fire}))$	
e	$(\text{Smoke} \rightarrow \text{Fire}) \rightarrow$ $((\text{Smoke} \wedge \text{Heat}) \rightarrow$ $\text{Fire})$	
f	$(\text{Fire} \rightarrow \text{Smoke}) \wedge$ $\text{Fire} \wedge \neg \text{Smoke}$	

Part 4 – Normal Forms (~10 min)

4.1 Convert the following expressions into **Conjunctive Normal Form (CNF)**:

(i) $\neg(s \rightarrow \neg p) \wedge (p \rightarrow \neg r)$

(ii) $p \leftrightarrow (\neg q \vee (q \rightarrow \neg p))$

(iii) $\neg(p \rightarrow (\neg q \vee (r \leftrightarrow \neg p)))$

4.2 Convert the following expression into **Disjunctive Normal Form (DNF)**:

$(s \rightarrow \neg(\neg r \rightarrow q)) \rightarrow p$

Part 5 – Python: Truth Tables (~10 min)

Open `logic_exercises.ipynb` and go to **Exercise 5**.

You are given a helper that evaluates a formula for one assignment. Your tasks:

5.1 Complete `truth_table(variables, formula_fn)` to print all rows.

5.2 Use it to verify your answers from Part 1.2 (the tautology) and Part 3 (validity table).

5.3 Use it to **count** the number of satisfying models for the formulas in Part 2.

Part 6 – Python: Resolution (~8 min)

In **Exercise 6** of the notebook you will implement a simple resolution checker.

6.1 A clause is a set of literals like `{ 'P', '¬Q', 'R' }`. Complete `resolve(c1, c2)` which returns the resolvent of two clauses (or `None` if they cannot be resolved).

6.2 Complete `pl_resolution(kb_clauses, goal_clause)` which tries to derive a contradiction from the KB + negated goal.

6.3 Use it to verify the unicorn problem from the worksheet: - If the unicorn is mystical $\rightarrow$ immortal; if not mystical $\rightarrow$ mortal mammal - Immortal or mammal $\rightarrow$ has horn; has horn $\rightarrow$ magical

Show that **the unicorn is magical**.

Part 7 – First-Order Logic: Formula Matching (~5 min)

Match each FOL formula to its correct English sentence. Write the letter next to the number.

#	Formula	Answer
1	$\exists x. (\text{big}(x) \wedge \text{cat}(x))$	
2	$\exists x. (\text{big}(x) \rightarrow \text{cat}(x))$	
3	$\exists x. (\text{big}(x) \vee \text{cat}(x))$	
4	$\forall x. (\text{big}(x) \wedge \text{cat}(x))$	

#	Formula	Answer
5	$\forall x. (\text{big}(x) \rightarrow \text{cat}(x))$	
6	$\forall x. (\text{big}(x) \vee \text{cat}(x))$	
7	$\forall x. (\text{big}(x) \vee \neg \text{cat}(x))$	

(**A**) All cats are big. (**B**) There exists a small thing or a cat. (**C**) All big things are cats. (**D**) All things are big cats. (**E**) All cats are small. (**F**) There exists a big cat. (**G**) All small things are cats. (**H**) There exists a big thing or a cat.

Part 8 – FOL Translation Critique (~5 min)

For each sentence, decide whether the FOL formula is a **correct** translation. If not, describe the mistake.

(a) “Every apartment in Munich is cheaper than some apartment in Starnberg.”

$$\forall x. (\text{App}(x) \wedge \text{In}(x, \text{Munich}) \rightarrow \exists y. (\text{App}(y) \wedge \text{In}(y, \text{Starnberg}) \rightarrow \text{Rent}(x) < \text{Rent}(y)))$$

(b) “There is exactly one apartment in Starnberg with rent below 500€.”

$$\exists x. (\text{App}(x) \wedge \text{In}(x, \text{Starnberg}) \wedge \forall y. (\text{App}(y) \wedge \text{In}(y, \text{Starnberg}) \wedge \text{Rent}(y) < 500 \rightarrow x = y))$$

(c) “If some apartment is more expensive than any apartment in Munich, it must be in Starnberg.”

$$\forall x. (\text{App}(x) \wedge \forall y. (\text{App}(y) \wedge \text{In}(y, \text{Munich}) \wedge \text{Rent}(x) > \text{Rent}(y)) \rightarrow \text{In}(x, \text{Starnberg}))$$

Good luck!